1. Executive Summary

Arch Systems, LLC is pleased to submit recommendations for developing a comprehensive Artificial Intelligence (AI) Action Plan that aligns with Enterprise Architecture (EA) principles, promotes cross-agency collaboration, and ensures rapid AI adoption while avoiding vendor lock-in and cost escalations through open-source solutions.

AI has the potential to improve decision-making, automate workflows, and enhance operational efficiency. However, AI must be implemented with agility, governance, and interoperability in mind to ensure scalability, security, and cost-effectiveness. This document offers actionable insights into breaking down silos, fostering AI-driven innovation, standardizing AI infrastructure, and ensuring that AI aligns with public sector federal government missions. Figure 1 illustrates the intelligent Human Centered Design (iHCD) AI lifecycle management platform.

Figure 1: Intelligent Human-Centered Design (iHCD) AI Lifecycle Management Platform

<u>The Cost-Effectiveness of Small Businesses in AI Innovation</u>

Small businesses like Arch Systems, LLC, consistently demonstrate that they can deliver high-quality AI solutions more efficiently and cost-effectively than larger enterprises. Small businesses are agile, responsive, and driven by innovation, unlike large corporations, which often encounter bureaucratic delays, high overhead costs, and rigid processes.

Over the past year, Arch Systems, LLC has successfully deployed over 20+ AI projects for the Administration for Children and Families (ACF) - a feat that larger businesses typically take years to accomplish. This success aligns with the Department of General Effectiveness (DOGE) principles, emphasizing efficiency, adaptability, and mission-driven AI deployment. Our lean, highly skilled teams provide customized AI solutions without unnecessary cost increases, ensuring that taxpayer dollars are utilized effectively. By leveraging open-source AI frameworks, cross-agency collaboration, and rapid development cycles, we deliver AI solutions that are not only cutting-edge but also fiscally responsible.

For the 2025 Disruptive Tech Awards, Arch is an Awardee for 2025!
ACF Tech's pioneering AI implementation for the Office of Head Start set a new precedent by publishing open-source code, marking a first for the agency. This initiative demonstrated a commitment to transparency and collaboration, enabling other federal agencies, researchers, and developers to learn from and build upon ACF's innovative work. By utilizing large language models (LLMs) to automate the tagging and analysis of public comments, the initiative streamlined a previously manual process, significantly reducing review times while enhancing accuracy. This effort not only modernized internal operations but also established a model for leveraging AI in public service, fostering a culture of openness, and advancing federal innovation standards across government agencies.

For the 2025 FXPElev8 Awardee - Recognizing High-Performing Federal Programs, Arch is an Awardee for 2025.

Innovative Partnerships	Empowerment & Accountability	Agility & Responsiveness
• Launched **Flexible Data Analytics & Infrastructure Support** program • Established **first-ever ACF Tech Data Surge Team** • Enables rapid, strategic data analysis & infrastructure projects	• Empowers **2,000 public servants** with AI-driven insights • Enhances leadership, partnerships, & resource delivery • Drives initiatives for **disaster preparedness, equity, & grant support**	• **25+** AI Proof of Concepts and Pilots created in less than 18 months • Rapid data-driven decision-making • Strengthens ability to respond to emergent agency needs

Table 1 outlines the bio of Dr. Alain C Briancon, our Executive AI Innovations Officer.

Table 1: Bio of Dr. Alain C. Briancon

Bio of Alain Briancon
 Dr. Alain C. Briancon is a distinguished technology executive with over 30 years of leadership experience as a VP, CTO, and CEO across large enterprises, medium-sized businesses, and innovative startups. He holds a Ph.D. and M.S. in Electrical Engineering and Computer Science from MIT, both earned with a perfect 5.0 GPA, and a Master's degree from Supelec, Paris, where he graduated as a valedictorian. With an impressive portfolio of 90 patents, including 29 in AI and machine learning, Alain has spearheaded innovations that significantly enhanced business performance, such as an AI-powered pricing system that managed $350 million for Kantar Profiles and an AI-based GDP forecasting model adopted by the Bank of Canada. His expertise spans AI, big data, IoT, security, and digital transformation, driving strategic initiatives like an AI approach to automotive loan adjudication that boosted volumes by $1 billion and a churn reduction model for Verizon's 125 million customers. As CTO at Cerebri AI, he led the development of real-time AI-driven platforms that optimized decision-making and reduced costs for Fortune 500 companies. Alain's ability to transform businesses through AI and advanced technologies and his strategic leadership and team-building skills make him a recognized authority in leveraging innovation to drive growth and operational excellence. Alain has participated as a reviewer to many NSF SBIR panels.

2. Aligning AI with Enterprise Architecture (EA)

To ensure AI integrates seamlessly into federal IT ecosystems, it must align with Enterprise Architecture (EA) principles, focusing on interoperability, cost control, security, and scalability. Table 2 describes AI-EA integration strategies.

Table 2: AI-Enterprise Architecture (EA) Integration Strategies

AI-EA Integration Strategies	Description
Open-Source AI Frameworks	Adopt open-source AI models (e.g., TensorFlow, PyTorch) to prevent vendor lock-in and control costs.
API-Driven AI Architecture	Develop standardized AI APIs to ensure cross-agency interoperability and data integration.
Hybrid Cloud & Edge AI	Implement scalable AI infrastructure that balances on-premises, edge computing, and cloud AI services.
Microservices for AI	Use containerized AI models (e.g., Kubernetes, Docker) for modular, flexible AI deployments.
Zero Trust AI Security	Embed security-first principles into AI workflows using federal cybersecurity guidelines (e.g., NIST, FedRAMP).

Aligning AI with EA frameworks ensures that AI deployments remain modular, adaptable, and cost-effective.

3. Promoting Cross-Agency Collaboration & Removing Silos

Government agencies often struggle with data silos, duplicative AI initiatives, and lack of interoperability. Arch proposes a federated AI collaboration model that ensures seamless data sharing and interoperability across agencies. Figure 2 illustrates key components of AI governance and collaboration.

Key Recommendations: By breaking down AI silos, government agencies can accelerate AI adoption while maximizing resource utilization.

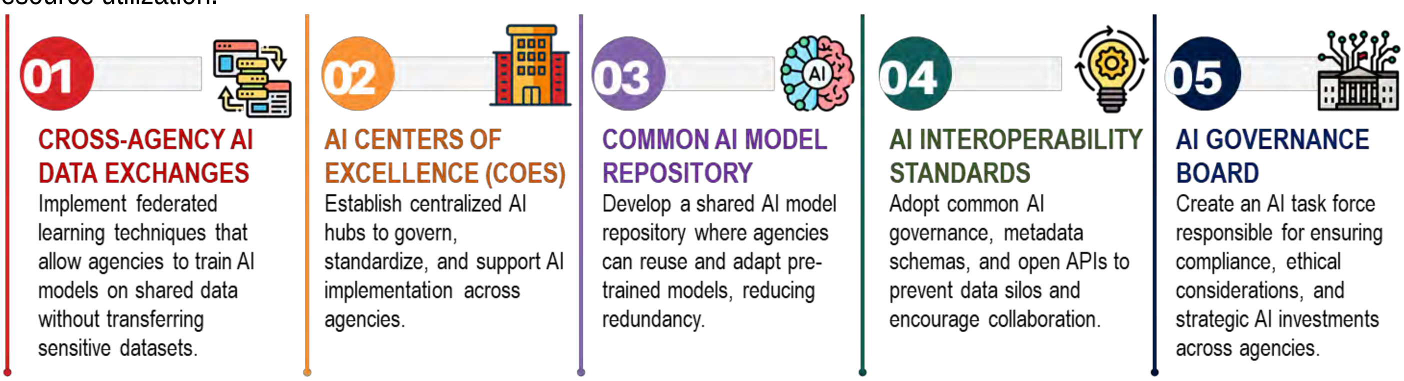

Figure 2: Key Components of AI Governance and Collaboration

4. Agility & Rapid AI Adoption: Pilot Programs & AI Platforms

To mitigate risks and ensure AI is thoroughly tested before full-scale deployment, Arch recommends a "fail fast, learn fast" approach through pilot AI programs and rapid prototyping platforms. Table 3 describes AI acceleration strategies and implementation approaches.

Table 3: AI Acceleration Strategies and Implementation Approaches

AI Acceleration Strategy	Implementation Approach
Regulatory AI Sandboxes	Establish AI testing environments with **controlled risk exposure for experimental AI applications**.
Open-Source AI Pipelines	Leverage open-source AI tools (e.g., MLflow, Hugging Face) to rapidly train, test, and deploy AI models.

AI Acceleration Strategy	Implementation Approach
Low-Code/No-Code AI Tools	Provide agencies with drag-and-drop AI automation platforms to promote non-technical AI adoption.
AI DevOps & MLOps	Automate AI development, model monitoring, performance logging, and deployment pipelines for faster iteration cycles.
Pilot AI Models with Open Data	Use publicly available datasets (e.g., data.gov) to test AI capabilities before production rollout.

5. Platforms for Idea Management & Continuous AI Innovation

To sustain long-term AI innovation, organizations need dedicated AI knowledge-sharing platforms.

Recommended AI Innovation Strategies: All of this can be accomplished within our Platform

- Open Innovation Challenges: Launch AI competitions and hackathons to crowdsource solutions from experts, academia, and industry partners.
- AI Knowledge Hubs: Establish a federal AI repository where agencies can share AI best practices, governance, data dictionaries, models, and datasets.
- AI Feedback & Monitoring Platforms: Implement real-time AI performance tracking systems to enhance transparency and accountability.
- Crowdsourced AI Solutions: Develop open-source AI incubators to empower federal employees and stakeholders to contribute AI-driven ideas.

6. Data & Infrastructure for AI Scalability

AI adoption requires robust, scalable, and cost-efficient infrastructures. Table 4 outlines the recommended infrastructure components for AI implementation.

Table 4: Recommended Infrastructure Components for AI Implementation

Infrastructure Component	Recommendation
Open-Source Data Lakes	Use open-source data lake platforms (e.g., Apache Hadoop, Delta Lake) for cost-efficient AI model training.
Federated Learning for Secure AI	Train AI models across agencies without centralizing sensitive data to enhance privacy.
Hybrid AI Compute Resources	Balance AI workloads across on-premises, edge, and cloud environments for efficiency.
Standardized AI APIs	Implement common API frameworks for seamless AI integration and scalability.

Adopting open-source, standardized AI architecture reduces costs and promotes flexible, scalable AI adoption.

7. Workforce & Change Management for AI Adoption

AI requires organizational and workforce transformation.

- AI Literacy for Government Employees: Establish mandatory AI training programs for federal employees.
- AI Augmented Workflows: Use AI-powered tools to enhance, not replace, human expertise.
- AI Ethics & Bias Training: Implement AI fairness and bias mitigation workshops.
- AI Workforce Development: Create cross-functional AI teams to ensure AI is adopted at all levels of government operations.

8. Security, Compliance, & AI Governance

AI security must be a top priority to prevent misuse. Table 5 outlines AI security best practices and implementation approach.

Table 5: AI Security Best Practices and Implementation Approach

AI Security Best Practices	Implementation Approach
Zero Trust AI Models	Require continuous authentication for **AI access and decision-making**.
Explainable AI (XAI)	Ensure AI models are **transparent and auditable**.
AI Bias & Ethics Audits	Mandate **regular AI fairness audits** to prevent **algorithmic discrimination**.
Secure AI APIs	Implement **strict access controls for AI-powered systems**.

AI deployments must adhere to NIST, GDPR, and federal AI ethics guidelines.

9. Global AI Cooperation & Benchmarking

- Adopt International AI Standards: Collaborate with ISO, IEEE, and NIST for AI regulation alignment.
- Participate in AI Diplomacy: Engage in global AI research and policy-sharing initiatives.
- Agencies can ensure robust, future-proof AI deployments by leveraging global AI best practices.

10. Conclusion & Call to Action

Arch Systems, LLC, advocates for AI strategies that emphasize open-source adoption, interoperability, and cost-efficiency. We welcome collaboration with federal agencies to shape AI implementation strategies.